

18-879 – Project Update: Extracellular Stimulation in Jaxley

Marco Christiani April 21, 2026

1. Problem Statement

1.1. System and Experimental Setting

Neurostimulation technique. We consider transcutaneous and implanted electrical neurostimulation, including deep brain stimulation (DBS) and spinal cord stimulation (SCS). In these modalities, one or more electrodes are placed outside the target neural tissue and deliver pulsed current waveforms. The resulting extracellular electric field passively drives transmembrane currents via spatial gradients in the potential – a mechanism captured by the classical activating function formalism. Waveform parameter selection (pulse width, shape, amplitude, frequency, polarity) determines therapeutic outcome, side-effect profile, and device battery life, yet is largely empirical in clinical practice. The design space is combinatorially large and neural response is highly nonlinear, making principled optimization intractable without fast, differentiable simulation. Examples of concrete downstream problem statements are further provided below.

Data and model. All data in this project are simulation outputs from a biophysically detailed in-silico model. The neural substrate is a multi-compartment Hodgkin-Huxley (HH) cable implemented in Jaxley, a JAX-based differentiable neuron simulator. The extracellular field is modeled analytically as a point-source monopole in an infinite, homogeneous, isotropic medium (conductivity $\sigma = 0.3$ S/m). There are no biological recordings; the simulator is the ground truth. Morphological parameters (cable length, fiber radius, electrode distance) and stimulation parameters (pulse width, waveform shape, frequency) constitute the experimental variables.

Problem type. The primary task is offline: given a fixed neural geometry and electrode placement, find the stimulation waveform that achieves a target neural response (e.g., action potential initiation) at minimum delivered charge. A secondary task is data generation: sweep the parameter space to characterize how neural response varies across geometries and waveform types.

1.2. Formal Problem Definition

Let a stimulation waveform be parameterized as a discrete sequence $w \in \mathbb{R}^T$ (electrode current in μA at each timestep). Let $\theta \in \Theta$ denote the biological and geometric parameters of a fixed experimental configuration: fiber radius a , cable length L , electrode distance d , tissue conductivity σ , and channel kinetics.

Simulation forward model. The simulator F maps a waveform to a set of response features:

$$F(w; \theta) = (\text{spiked}, \text{latency}, \text{spike_count}, \text{firing_rate})$$

Internally, F solves the standard multi-compartment cable equation with extracellular coupling:

$$c_m \frac{dv_i}{dt} = -I_{\text{ion}}(v_i, \mathbf{s}_i) + \sum_j g_{ij}(v_j - v_i) + \sum_j g_{ij}(\phi_j(t) - \phi_i(t))$$

where v_i is the membrane potential (mV) of compartment i , c_m is the specific membrane capacitance, $I_{\text{ion}}(v_i, \mathbf{s}_i)$ is the total membrane ionic current density governed by gating states $\mathbf{s}_i$, g_{ij} are axial coupling coefficients between adjacent compartments, r_j is the distance from the electrode to compartment j , and $\phi_j(t)$ is the extracellular potential at compartment j due to a point-source monopole electrode in an infinite homogeneous medium:

$$\phi_j(t) = \frac{w(t) \cdot 10^3}{4\pi\sigma r_j} \quad [\text{mV}]$$

with w in μA , σ in S/m, and r_j in μm (the base units cancel to V; the 10^3 converts to mV for consistency with v).

The extracellular term acts as a forcing: the spatial gradient of ϕ across compartments drives transmembrane current even in the absence of intracellular injection.

Data generation. For a fixed waveform family (e.g., rectangular pulses parameterized by amplitude A and pulse width t_p), define the threshold amplitude:

$$A^*(t_p; \theta) = \min\{A \geq 0 \mid F(w_{A,t_p}; \theta) \cdot \text{spiked} = 1\}$$

The goal is to characterize A^* and the associated response features across the full configuration space $\mathcal{P} \times \mathcal{W} \times \mathcal{F} \times \Theta$ (pulse widths, waveform types, frequencies, and geometric/biophysical parameters), producing a labeled dataset $\mathcal{D}$.

Waveform optimization. For a free-form waveform $w \in \mathbb{R}^T$, minimize delivered charge subject to achieving neural activation:

$$w^* = \arg \min_{w \in \mathcal{C}} \|w\|_1 \cdot \Delta t \quad \text{subject to} \quad \hat{F}(w; \theta) \geq \tau$$

where Δt is the simulation timestep (so $\|w\|_1 \cdot \Delta t$ is total delivered charge), $\hat{F}$ is a continuous relaxation of the binary activation indicator, τ is an activation threshold, and $\mathcal{C}$ encodes safety constraints (amplitude bounds, charge balance $\sum_t w_t = 0$ for biphasic waveforms).

1.3. Downstream Problem Statements

The dataset $\mathcal{D}$ and gradient access from the optimization formulation above support two concrete downstream problems, stated here independently of approach.

Energy efficiency. Given a target response (e.g., single spike in fiber θ), find the minimum-charge waveform. This is the optimization above evaluated across the θ grid to characterize how waveform shape interacts with fiber geometry at threshold.

Selectivity. In practice, the tissue near a stimulation electrode contains neurons with different geometries – for example, target neurons near the electrode and non-target axons passing through the stimulation field at greater distance or with different fiber radius. Given two neural configurations θ_1 (target) and θ_2 (non-target), find a waveform that activates θ_1 while sparing θ_2 :

$$w^* = \arg \min_{w \in \mathcal{C}} \|w\|_1 \cdot \Delta t + \lambda \cdot \tilde{F}(w; \theta_2) \quad \text{subject to} \quad \tilde{F}(w; \theta_1) \geq \tau$$

where Δt is the simulation timestep (so $\|w\|_1 \cdot \Delta t$ is total delivered charge), $\tilde{F}$ is the smooth activation surrogate defined above, τ is the activation threshold, and λ controls the tradeoff between charge minimization and sparing the non-target fiber. The penalty formulation (rather than a hard constraint on θ_2) keeps the problem smooth and allows tracing the selectivity–efficiency Pareto frontier by varying λ .

2. Approach and Progress to Date

Milestone 1 – ECS pipeline (complete). Implemented the extracellular stimulation formalism as a fully differentiable, JAX-native module that couples to Jaxley through its public `data_stimulate` API without modifying Jaxley source.

Jaxley does not store raw axial conductances g_{ij} . Instead, it pre-normalizes them into a voltage diffusion operator G (units: 1/ms) with specific capacitance c_m , membrane area, and unit conversions already incorporated, so that the cable equation it solves internally is:

$$\frac{dv_i}{dt} = -\frac{I_{\text{ion}}(v_i, \mathbf{s}_i)}{c_m} + \frac{i_{\text{ext}}}{c_m} + [Gv]_i$$

The extracellular coupling from the problem statement, $\sum_j g_{ij}(\phi_j - \phi_i)$, enters through the same operator by linearity: $G(v + \phi_e) = Gv + G\phi_e$. The Gv term is Jaxley’s standard axial dynamics.

The extracellular forcing we need to inject is $[G\phi_e]_i$ (mV/ms), but Jaxley’s public stimulus API accepts current in nA. We therefore encode:

$$i_{\text{ecs}}[\text{nA}] = c_m \cdot (G\phi_e) \cdot \text{area} / 10^5$$

Jaxley internally converts $i_{\text{nA}} \rightarrow i_{\text{ext}}[\mu\text{A}/\text{cm}^2] = i_{\text{nA}} / \text{area} \cdot 10^5$, then adds i_{ext}/c_m to dv/dt , recovering $G\phi_e$ exactly. The c_m and area factors cancel in the round-trip – they are an encoding artifact of working through Jaxley’s public API, not part of the physics. This round-trip is verified to $\sim 10^{-13}$ relative error in float64.

The operator G was reconstructed from Jaxley’s internal axial conductances, including reverse-engineering a five-type edge system for branchpoint pseudo-nodes eliminated via Gaussian substitution. Validation:

1. Agreement with Jaxley’s internal `_voltage_vectorfield` to $\sim 10^{-13}$ relative error
2. $O(dx^2)$ convergence against the analytical activating function for a straight cable (error ratio $\sim 4 \times$ when doubling compartment count)

Gradients with respect to waveform shape, electrode position, and tissue conductivity are verified end-to-end through `jx.integrate`.

Milestone 2 – TPU-scale infrastructure (complete). Built JAX device mesh sharding, xarray/Zarr labeled storage with incremental append, and an experiment tracking stack. Cloud infrastructure is provisioned declaratively via OpenTofu including TPU instances, Cloud SQL Postgres, GCS artifact bucket, and a dedicated GCE tracking server.

Milestone 3 – Large-scale dataset (in progress). Executed a 1,296-configuration sweep over pulse width (9 values, 0.05–2.0 ms), waveform type (monophasic cathodic/anodic, biphasic), stimulation frequency (0, 100, 200, 500 Hz), electrode distance (25–200 μm), and fiber radius (5–20 μm). For each configuration, threshold amplitude was found by binary search and firing pattern features (spike count, ISI, firing rate) were recorded by re-simulation at threshold. These axes of variation were initially chosen as they have intuitive tradeoffs:

- pulse width: strength-duration tradeoff
- electrode distance: field attenuation and threshold increase
- polarity/waveform shape: activation asymmetry and charge efficiency
- frequency: repeated-response dynamics / firing patterns

Pending feedback on whether larger scale data is desirable, parameters can be added to the simulation but unclear what would be interesting from a neuroscience perspective. Moreover, the codebase has been engineered for scale far beyond what these experiments have demanded.

Milestone 4 – Gradient-based optimization (pending). The differentiability infrastructure is in place (verified in Milestone 1). Remaining work: (1) smooth surrogate for the binary spike indicator, (2) loss function and constraint formulation, (3) optimizer loop (possibly Adam or L-BFGS over free-form waveform), and (4) comparison of optimized waveform charge against the Milestone 3 rectangular-pulse baseline.

3. Addendum: Clarifying Motivation and Impact

The standard computational approach couples a volume conductor model (for the extracellular field) to the NEURON simulator (for the neural response). NEURON is CPU-bound and executes simulations serially, so parameter sweeps over waveform shape, geometry, and electrode placement scale linearly in wall-clock time. To work around this, researchers have used genetic algorithms to search over waveform shapes [1], [2], which require thousands of forward simulations and provide no gradient information, or trained surrogate neural networks on NEURON outputs [3], which are fast at inference but introduce approximation error and must be retrained when the model changes. Neither approach provides exact gradients through the biophysical simulation.

Jaxley [4] removes these bottlenecks: as a JAX-native simulator it supports automatic differentiation (exact gradients of any output with respect to any parameter), JIT compilation, GPU/TPU acceleration, and native batch parallelism via `vmap`. A single strength-duration sweep that requires serial binary searches in NEURON can be vectorized across the full parameter grid in one `vmap` call. More importantly, gradient access enables direct optimization of free-form waveforms – the Milestone 4 formulation below – without surrogate approximation or evolutionary search. This project extends Jaxley to extracellular stimulation, which it does not currently support, to bring these capabilities to the neurostimulation design problem.

These capabilities matter concretely for clinical neurostimulation. DBS parameter tuning is currently performed by a clinician manually adjusting settings over weeks of patient visits, guided by symptom observation rather than principled search. Spinal cord stimulation faces a similar bottleneck: waveform parameters that work for one electrode placement may fail after lead migration, requiring reprogramming. Fast, differentiable simulation could compress these design cycles from trial-and-error clinical sessions to automated computational sweeps – and gradient-based optimization could discover waveform shapes that no clinician would arrive at by manual tuning.

18-879 – Project Update 2: Extracellular Stimulation in Jaxley

Marco Christiani April 21, 2026

4. Progress Since Update 1

4.1. Implementation

Since the first update, we have:

- **Multi-electrode support.** Generalized the field model from a single electrode to N electrodes via superposition: $\phi_e(\mathbf{x}, t) = \sum_{i=1}^N \frac{I_i(t) \cdot 10^3}{4\pi\sigma\|\mathbf{x}-\mathbf{x}_i\|}$. We formulated an implementation that allows for full differentiation so we can support gradient-based optimization.
- **HH dynamics and experiment infrastructure.** Built user-friendly abstractions to encapsulate the full pipeline: module setup, operator construction, field computation, equivalent-current encoding, and Jaxley integration. Includes vectorized binary search for threshold finding via `jit(vmap(...))` and response feature extraction (spike detection, latency, firing rate).
- **Visualization.** Added rendering utilities (morphology + field potential heatmap + electrode markers, voltage animation, broken-axis plots) composable with Jaxley's `vis()`.
- **1,296-configuration sweep on TPU.** Swept pulse width (9 values), waveform type (3), frequency (4), electrode distance (4), and fiber radius (3) on a TPU instance.

4.2. Analytical Derivations

The first update received feedback that the relationship between our discrete operator G and the continuous cable equation was not sufficiently clear. We worked on developing more formal derivations for both the straight-cable and branching cases.

4.2.1. Straight cable: $G\phi_e$ recovers the activating function

Consider a function $\phi(x)$ sampled at uniformly spaced points x_{i-1}, x_i, x_{i+1} with spacing Δx . The forward and backward finite difference approximations of the first derivative at x_i are:

$$\phi'_+ = \frac{\phi_{i+1} - \phi_i}{\Delta x}, \quad \phi'_- = \frac{\phi_i - \phi_{i-1}}{\Delta x}$$

Taking the finite difference of these two approximations gives the second derivative:

$$\phi''_i \approx \frac{\phi'_+ - \phi'_-}{\Delta x} = \frac{\phi_{i-1} - 2\phi_i + \phi_{i+1}}{\Delta x^2}$$

This is the standard central difference stencil $[1, -2, 1]/\Delta x^2$, which approximates $d^2\phi/dx^2$ to $O(\Delta x^2)$.

The discrete diffusion operator G used in the cable equation has exactly this structure: for interior compartments of a uniform cable, $[G\phi]_i$ is proportional to $\phi_{i-1} - 2\phi_i + \phi_{i+1}$, i.e., it computes a scaled discrete second derivative. Applying G to the extracellular potential ϕ_e therefore approximates the classical activating function $f(x) \propto d^2\phi_e/dx^2$ [5]. We were also able to verify the $O(\Delta x^2)$ convergence numerically: doubling the number of compartments used in our model reduces the error by a factor of $\sim 4 \times$.

4.2.2. Branching morphologies

In the first update we were only considering structures that do not contain branches. Since then, we worked towards supporting branching structures and this has proven non-trivial. We have made progress by looking closely at Jaxley internals and attempting to back-out the underlying mathematics. The following represents our understanding thus far.

Jaxley introduces branchpoint pseudo-nodes at cable junctions to enforce Kirchhoff's current law. The full system has the block structure:

$$\begin{pmatrix} G_{RR} & G_{RB} \\ G_{BR} & G_{BB} \end{pmatrix} \begin{pmatrix} v_R \\ v_B \end{pmatrix} = \begin{pmatrix} [Gv]_R \\ [Gv]_B \end{pmatrix}$$

where R denotes real compartments and B denotes branchpoint nodes. Since branchpoint nodes carry no membrane and no capacitance, they satisfy the instantaneous constraint $[Gv]_B = 0$, giving:

$$v_B = -G_{BB}^{-1} G_{BR} v_R$$

Substituting into the real-compartment rows yields the Schur complement:

$$\tilde{G} = G_{RR} - G_{RB} G_{BB}^{-1} G_{BR}$$

so that $\tilde{G}v_R$ gives the axial contribution at real compartments only. Our implementation encodes this elimination when assembling from Jaxley's edge data, producing $\tilde{G} \in \mathbb{R}^{N_{\text{comp}} \times N_{\text{comp}}}$ directly. The same operator applies to extracellular forcing: $\tilde{G}\phi_e$ gives the activating function at real compartments only.

5. Next Priorities

1. **Formalization.** Highest priority item is completing and consolidating our mathematical formulations then aligning notation with our implementation and documentation. As this nears completion, it will be shared with the instructors as discussed along with relevant citations.
2. **PV and Pyr neuron models.** Jaxley does not appear to provide PV or Pyr cell models. We will implement custom channel kinetics for these cell types to enable cell-type-specific stimulation studies.
3. **External baseline validation.** Compare our PV/Pyr simulation results against the NEURON-based simulator used in the course to validate that our Jaxley+ECS pipeline reproduces established results.
4. **Analytical validation of branching.** Derive closed-form activating functions for simple branching geometries (Y-junction) and verify $\tilde{G}\phi_e$ convergence, extending the straight-cable derivation above. This should be further corroborated by empirical evidence. This may become a stretch goal depending on how quickly the higher priority items can be accomplished, as this is not a central component.
5. **(Stretch goal) Refined waveform parameterization.** Develop a charge-balanced waveform family parameterized by (t_+, t_-, A_+, f) where the remaining variables (e.g., A_-) are solved from charge-balance and timing constraints. This should yield a more representative search space for the application domain.
6. **(Stretch goal) Gradient-based waveform optimization.** With validated branching morphologies and realistic neuron models, apply the differentiable pipeline to one of the optimizations problem previously proposed (in the proposal and initial update).

18-879 – Project Update 3: Extracellular Stimulation in Jaxley

Marco Christiani April 21, 2026

6. Progress Since Update 2

6.1. Formalization

A draft mathematical formulation document has been shared with the instructors as discussed. It remains an active work-in-progress: references are still being confirmed, some derivation steps are being solidified, and overall presentation is ongoing. This has taken the majority of my time so far, and I anticipate will take up a considerable amount of the remaining time.

6.2. Operator Simplification

I identified a meaningful simplification to the construction of $\tilde{G}$. The previous implementation replicated Jaxley's internal logic to assemble the sparse operator from scratch. I found some private Jaxley internals for doing the necessary computation, so our construction now delegates to it rather than duplicating it. The result is semantically identical (the same $\tilde{G} \in \mathbb{R}^{N_{\text{comp}} \times N_{\text{comp}}}$ via the same Schur complement elimination) but is considerably less complex, more concise, less likely to drift out of sync with upstream changes, and depends less on me having as-thorough of an understanding of Jaxley's internal modeling.

6.3. BBP L2/3 Neuron Models

I have implemented Pyr and PV cell factory functions for the BBP L2/3 reference models (L23_PC_cADpyr229 and L23_LBC_cNAC187). Channel kinetics are taken from existing Jaxley-compatible implementations of the L5PC channel set. Conductance values and passive parameters are extracted from the BBP biophysics definitions included in previous assignments and applied per section group (soma, axon, apical, basal).

Morphology loading proved non-trivial. The BBP source morphologies represent the soma as a 2D contour, and it seems this cannot be encoded in SWC. The conversion tool emits soma points with zero radius as a consequence, which causes Jaxley to compute degenerate zero-area compartments and produce NaN voltages at the first integration step. This required patching the converted files before use. Additionally, long branches are subdivided on load to bring Jaxley's segment count closer to what NEURON's automatic segmentation heuristic would produce. Ideally, I find simplifications that can be made so verification is possible without inter-framework adaptations causing a lack of fidelity.

Calcium signaling required delicate channel insertion order: the Nernst reversal potential for calcium must be established before calcium-carrying channels, which must precede the calcium pump and calcium-activated potassium channels. Incorrect ordering produces uninitialized state references at integration time. I am hesitant on my understanding of this but am getting more comfortable as I work with the concepts.

The implemented models pass assembly checks against the source biophysics, integration smoke tests, resting potential range checks, and spike elicitation tests under somatic current injection.

6.4. NEURON Parity Validation (In Progress)

Work toward external validation against the NEURON-based simulator used in the course is underway. The approach is to inject an identical somatic step current into both simulators and compare voltage traces and spike metrics (resting potential, spike count, first-spike latency, AP amplitude).

Several compatibility issues required resolution before a meaningful comparison is possible. The course-provided NEURON models enable axon myelination by default, which our Jaxley implementation does not support, so I disable it in the NEURON reference to keep the comparison on equal footing. The Jaxley-compatible channel implementations apply a fixed Q10 temperature correction, requiring the NEURON reference to be run at a matching temperature. Compartmentalization also differs

between the two frameworks, I attempt to control this through branch-splitting on the Jaxley side to approximate NEURON's segmentation heuristic.

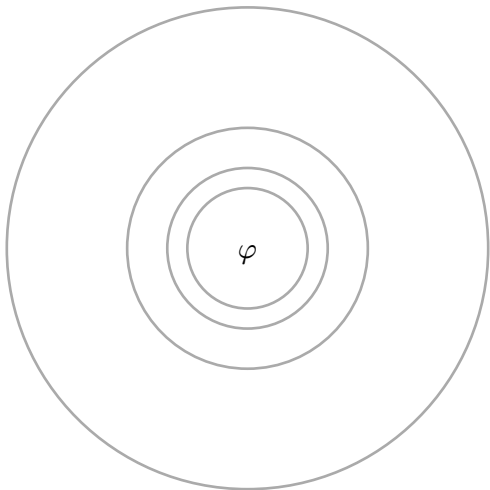
With these steps in place, comparison is possible, and work is ongoing. The results are not yet in agreement, though qualitative features of the response appear broadly consistent. The remaining discrepancies are under investigation and possible sources include compartmentalization differences, integration scheme differences, and parameter mapping subtleties between the original HOC definitions and the Jaxley channel implementations.

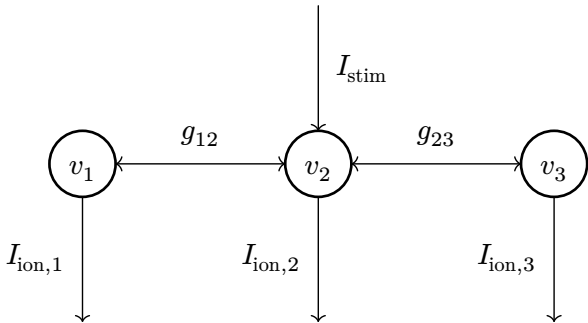
7. Next Priorities

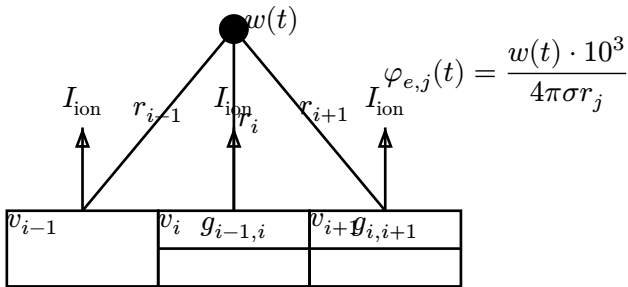
Given the remaining timeline and the complexity of the validation work, the primary focus going forward is achieving parity with NEURON alongside the formalization with proper citations. Other items from the previous update are carried forward as stretch goals.

1. **NEURON parity.** Characterize and resolve the remaining discrepancies between the Jaxley and NEURON simulations of the L23 Pyr model. If full suprathreshold parity proves intractable, establish agreement at minimum on passive membrane properties and resting potential as a baseline.
2. **ECS stimulation with realistic morphologies.** Apply the full extracellular stimulation pipeline to the BBP Pyr and PV cells and characterize threshold currents and activation patterns for each cell type.
3. **(Stretch) Formalization.** Continue refining the derivation document: confirm references, finalize remaining derivation steps, and incorporate instructor feedback.
4. **(Stretch) Analytical validation of branching.** Derive closed-form activating functions for a simple branching geometry and verify $\tilde{G}\phi_e$ convergence, extending the straight-cable derivation from Update 2.
5. **(Stretch) Gradient-based waveform optimization.** With validated cell models, apply the differentiable pipeline to waveform optimization as originally proposed.

0.07957747154594767







$$G(v + \varphi_e) = Gv + G\varphi_e$$

– Scratch Work

Marco Christiani April 21, 2026

For a smooth function $\phi(x)$,

$$\phi(x+h) = \phi(x) + h\phi'(x) + \frac{h^2}{2}\phi''(x) + \frac{h^3}{6}\phi'''(x) + \frac{h^4}{24}\phi''''(x) + O(h^5)$$

At grid point x_i , let $h = \Delta x$. Then:

$$\phi_{i+1} = \phi_i + \Delta x \phi'_i + \frac{\Delta x^2}{2}\phi''_i + \frac{\Delta x^3}{6}\phi'''_i + \frac{\Delta x^4}{24}\phi''''_i + O(\Delta x^5)$$

and

$$\phi_{i-1} = \phi_i - \Delta x \phi'_i + \frac{\Delta x^2}{2}\phi''_i - \frac{\Delta x^3}{6}\phi'''_i + \frac{\Delta x^4}{24}\phi''''_i + O(\Delta x^5)$$

Adding the two expansions gives

$$\phi_{i+1} + \phi_{i-1} = 2\phi_i + \Delta x^2\phi''_i + \frac{\Delta x^4}{12}\phi''''_i + O(\Delta x^6)$$

So the odd-derivative terms cancel, and addition isolates the even derivatives.

Subtracting the two expansions gives

$$\phi_{i+1} - \phi_{i-1} = 2\Delta x \phi'_i + \frac{\Delta x^3}{3}\phi'''_i + O(\Delta x^5)$$

So the even-derivative terms cancel, and subtraction isolates the odd derivatives.

The symmetry is why centered stencils work well.

Bibliography

- [1] W. M. Grill, “Model-based analysis and design of waveforms for efficient neural stimulation,” *Progress in Brain Research*, vol. 222, pp. 147–162, 2015.
- [2] A. Wongsarnpigoon and W. M. Grill, “Energy-efficient waveform shapes for neural stimulation revealed with a genetic algorithm,” *Journal of Neural Engineering*, vol. 7, no. 4, p. 46009, 2010.
- [3] N. Méliissano and others, “Highly efficient modeling and optimization of neural fiber responses to electrical stimulation,” *Nature Communications*, vol. 15, p. 7524, 2024.
- [4] M. Deistler and others, “Jaxley: differentiable simulation enables large-scale training of detailed biophysical models of neural dynamics,” *Nature Methods*, 2025.
- [5] F. Rattay, “Analysis of models for external stimulation of axons,” *IEEE Transactions on Biomedical Engineering*, no. 10, pp. 974–977, 1986.